

Gamified Learning for Accessibility: Teaching Through Play at Limbitless Solutions

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Abstract

How can gamification and interactive learning revolutionize computer graphics education while making a tangible social impact? At Limbitless Solutions, an interdisciplinary team of 61 undergraduate students from six UCF colleges collaborates to develop adaptive video games that help individuals with limb differences and neuromuscular conditions learn to use bionic prosthetics and hands-free wheelchairs. This talk explores how serious games, interactive techniques, and accessibility-driven design foster engagement, empathy, and innovation in CG classrooms and beyond.

CCS Concepts

• **Applied computing** → **Fine arts; Interactive learning environments**; • **Human-centered computing** → **User interface design; Accessibility design and evaluation methods**.

Keywords

Adaptive or Assistive Technology, Art, Design, Education-Undergraduate, Games, Maker

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1 Introduction

Traditional lecture-based methods often struggle to engage students in complex subjects like computer graphics and animation. Experiential learning through serious games provides a compelling alternative, immersing students in problem-solving, human-centered design, and interdisciplinary collaboration.

At Limbitless Solutions, University of Central Florida (UCF) undergraduate students from six UCF colleges collaborate to develop

real-world training games that blend accessibility, game development, and interactive media. These projects serve a dual purpose:

- Empowering individuals with limb differences and ALS patients through gamified assistive technology training, increasing confidence and independence.
- Teaching undergraduate students to design with empathy and purpose, ensuring emerging CG professionals understand accessibility-driven design.

Engaging students in developing these tools, we not only provide life-changing accessibility solutions but also cultivate the next generation of designers, engineers, and digital artists who prioritize inclusivity in their work.

2 The Power of Gamified Learning

Learning through play is a driver of skill acquisition, especially for individuals adapting to assistive technology. Limbitless Solutions' interdisciplinary approach brings together students from animation, game design, biomedical engineering, and health sciences to create interactive training games. These projects not only advance accessibility but also redefine CG education, engaging students in real-world problem-solving while fostering empathy-driven design.

Gamification techniques such as reward-based progression, real-time feedback, and adaptive challenges enable users to improve their motor skills and familiarity with assistive technology incrementally. This hands-on, experiential learning approach has proven more effective than traditional methods of teaching motor skill adaptation.

For instance, Project Xavier allows individuals to control a wheelchair using only EMG signals from their facial muscles, restoring mobility and independence through adaptive gaming mechanics.

3 Methodology

The training games at Limbitless Solutions integrate the following:

- Gamification techniques: Reward-based progression, real-time feedback, and adaptive challenges designed to motivate trainees and student developers.
- Adaptive controllers: EMG-based controllers are used for ALS patients and limb-different users with different limbs to gain independence and confidence in using assistive technology.

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- User-centered design: Iterative testing with individuals with disabilities to refine mechanics and accessibility features, ensuring these tools meet real user needs.

This interdisciplinary approach allows students to develop cutting-edge solutions at the intersection of computer graphics, healthcare, and accessibility. By working directly with users, students gain first-hand experience in designing technology that has both functional and emotional impact.

They learn to anticipate challenges, refine real-world usability, and, most importantly, see their designs empower individuals meaningfully. This hands-on experience reinforces technical skills and the ethical and human-centered responsibility that comes with designing for accessibility.

4 Real-World Applications and Impact

Our bionic training games use interactive storytelling and accessibility-focused mechanics to provide engaging, structured learning experiences. These games simulate real-world scenarios that enhance fine motor control, reaction time, and adaptability.

Through research, we have found that gamified accessibility training reduces the learning curve for assistive technology adoption, increasing user confidence and independence. Project Xavier, for example, enables individuals to control a wheelchair using only EMG signals from their facial muscles, allowing hands-free mobility.

In the classroom, students benefit from direct exposure to real-world problem-solving, gaining industry-relevant experience that positions them as leaders in accessible technology development.

5 Educational Framework and Classroom Integration

Educators can foster engagement, collaboration, and technical skill-building by embedding serious game development into CG curricula. This talk presents a scalable, adoptable model that educators can implement, showcasing how assignments rooted in game mechanics, interactive design, and accessibility can inspire students to apply their CG skills to real-world challenges.

Educators will learn:

- How to integrate real-world projects into their courses, providing students with hands-on learning experiences.
- Methods for teaching human-centered design in interactive media, animation, and digital art.
- The value of interdisciplinary collaboration brings together students from CG, animation, game design, and healthcare fields.

6 Takeaways

- How serious games and gamification enhance accessibility and skill acquisition.
- The power of interdisciplinary collaboration in CG education.
- Practical techniques for integrating interactive, experiential assignments into courses.
- How educators can inspire students to develop empathy-driven design solutions

7 Conclusion

Bridging computer graphics, accessibility, and interactive storytelling, this talk challenges educators to rethink digital learning.

Our approach at Limbitless Solutions provides a framework for engaging, meaningful, and inclusive education—showcasing how technology can empower users while fostering the next generation of CG professionals and accessibility advocates.

This session will provide actionable strategies for integrating serious game development, accessibility, and interdisciplinary collaboration into CG curricula, ensuring students graduate with technical excellence and a commitment to inclusive design.

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